Haowen Yan

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BIOGRAPHY

I am Haowen Yan, a third-year student majoring in Computer Science and Technology at Sun Yat-Sen University. My research interests focus on **Embodied AI**, **Robotics**, **Reinforcement Learning**, **World Models**.

My long-term research goal is to develop robotic systems capable of operating in complex and hazardous environments, such as fire or earthquake scenarios, to assist in rescue missions and ultimately benefit society and our humanity.

EDUCATION

Sun Yat-Sen University — Guangdong, China
Bachelor in Computer Science — Sept 2023 – Present

- **GPA:** 4.3493/5.0; **Average Score:** 93.493/100; **Rank:** 3/353 (top 0.85%)
- **Course Grades:**
  Control Theory (100/100), Optimization Theory (99/100), Mathematics Analysis II (98/100), Machine Learning (98/100), Data Structures (98/100), Signals and Systems (97/100), Probability and Statistics (96/100), etc.
- **Self-Study Courses based on Interest:**
  Reinforcement Learning, Optimal Control, Robotics, etc.

PUBLICATIONS

***** indicates equal contribution.

- Tianxing Chen*, Zixuan Li*, Weijie Wan*, **Haowen Yan***, et al. "RoboDojo: A Unified Sim-and-Real Benchmark for Efficient Evaluation of Generalist Robot Manipulation Policies", **ICRA** Workshop, 2026.
- Tianxing Chen*, Yue Chen*, Zixuan Li*, Junyuan Tang*, Kailun Su*, Haoran Lu*, Weijie Wan*, Baijun Chen*, Songling Liu*, **Haowen Yan**, et al. "RoboDojo: A Unified Sim-and-Real Benchmark for Comprehensive Evaluation of Generalist Robot Manipulation Policies", arXiv preprint, 2026.

AWARDS & HONORS

Honors

- **2024 National Scholarship** (Ranked 1st out of 159 students in comprehensive evaluation)
- **2025 National Scholarship** (Ranked 8th out of 352 students in comprehensive evaluation)
- **2024 First-Class Academic Scholarship** (Awarded to top 8% students in comprehensive evaluation)
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Competition Awards

• China Undergraduate Operating System Design Competition	**National Third Prize**	2025.08
• China Undergraduate Mathematics Competition	**Provincial First Prize**	2024.11
• Shenzhen Cup Mathematical Modeling Competition	**Provincial Second Prize**	2024.12
• China Undergraduate Mathematical Modeling Competition	**Provincial Third Prize**	2024.10
• Mathematical Contest in Modeling, USA	**H Award**	2025.05
• MathorCup Mathematical Modeling Competition	**Provincial Second Prize**	2024.04

RESEARCH EXPERIENCE

RoboDojo: A Unified Sim-and-Real Benchmark for Efficient Evaluation of Generalist Robot Manipulation Policies 2026.03–2026.05

- **Introduction**: To improve the standardization and fairness of robot evaluation, we present RoboDojo, a unified sim-and-real benchmark for evaluating generalist robot manipulation policies.
- **Involvement**:
  - Designed a standardized real-robot evaluation platform, ensuring consistency and reproducibility of the evaluation environment across all aspects and details to facilitate large-scale benchmarking.
  - Trained mainstream policies (including π_0, $\pi_{0.5}$, ACT, etc.) and evaluated them on real robots in the RoboDojo platform.
  - Designed a web dashboard for live monitoring of evaluation progress and for viewing participants' evaluation video replays and related metadata.
  - Implemented an object reset routine on the real-robot evaluation platform to ensure items are placed at the same locations before each trial, guaranteeing fairness and consistency of evaluations.
- **Outcomes**:
  - The paper is currently under review.
  - The evaluation platform is currently applying for a national utility-model patent.

Ultra-Fast Model Predictive Controller (MPC) 2024.12–2025.12

- **Introduction**: Model Predictive Control (MPC) is a robust control method, but its high computational cost limits real-time deployment. We propose a hardware–software co-design approach to accelerate MPC and validate it through UAV experiments.
- **Involvement**:
  - Modeled UAV dynamics and verified state-space equations via MATLAB simulations.
  - Programmed the MPC on an FPGA and controlled UAV flight through ROS + Gazebo simulation.
  - Deployed a hardware-software co-accelerated MPC on a UAV and tested its performance.
- **Outcomes**:
  - This project awarded a provincial innovation grant.
  - Related paper is being prepared for submission.

Operating System Based on the RISC-V Instruction Set 2025.04–2025.08

- **Introduction**: Developed an OS on RISC-V, supporting up to 90 system calls; participated in the Operating System Design Competition.
- **Involvement**:
  - Participated in the overall architecture design of the operating system.
  - Implemented most process management related system calls.
  - Adapted and debugged low-level drivers.
- **Outcomes**:
  - Project won the National Third Prize in the 2025 China Undergraduate Operating System Design Competition.

Research on Teleoperation of A Robotic Arm 2026.01–2026.02

- **Introduction**: In many existing teleoperation systems, the robot motion is affected by the head-mounted display (HMD): even when the controller remains stationary, movement of the HMD alone can cause unintended robot motion.
- **Involvement**:
  - Implemented teleoperation of an existing robotic arm using Meta Quest 3S based on an incremental control scheme.
  - By introducing the initial pose of the HMD as a reference frame, this work eliminates the influence

of HMD motion, thereby improving system stability and control accuracy.

Research on Diffusion Models 2025.12–2026.01

- **Introduction**: Studied diffusion models and implemented DDPM, DDIM, and DPM-Solver (1st-, 2nd-, and 3rd-order) samplers, and improved sampling efficiency.
- **Involvement**:
  - Used Pytorch to implement DDPM, DDIM and DPM-Solver (1st/2nd/3rd-order) samplers.
  - Trained and tested them on the Smithsonian Butterflies subset dataset.
  - Improved DPM-Solver with v-prediction to accelerate sampling and enhance sample quality.

Research on Single-cycle CPU and 5-stage pipelined CPU (MIPS) 2024.12–2025.01

- **Introduction**: Designed and implemented a single-cycle CPU and a 5-stage pipelined CPU based on the MIPS architecture, covering the complete workflow from hardware design and verification to FPGA deployment.
- **Involvement**:
  - Designed a single-cycle CPU and a 5-stage pipelined CPU, with all hazards (data hazard, structural hazard, and control hazard) resolved in hardware.
  - Implemented both CPUs in Verilog, simulated and debugged them
  - Deployed CPUs on an FPGA development board, respectively.

SKILLS

English:	CET-4 (622), CET-6 (556)
Programming:	Python, Pytorch, C/C++, Matlab, Rust, Verilog
Software:	Linux, Mujoco, ROS, Gazebo, Git
Traits:	Innovative; diligent and detail-oriented; self-learner; persistent and hardworking